Data Semantics

Assignemnt 1 :

In the first step, I chose my favorite actress that I want to make queries about named “Jennifer Aniston.”

The first question we want to find out is :

1. Is she married?

Query in SPARQL:

```
PREFIX dbr: <http://dbpedia.org/resource/>

PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?spouse WHERE {

  dbr:Jennifer_Aniston dbo:spouse ?spouse .

}
```

Answer : Brad Pitt

2. How many films did she star in?

```
PREFIX dbr: <http://dbpedia.org/resource/>

PREFIX dbo: <http://dbpedia.org/ontology/>


SELECT (COUNT(?film) AS ?numFilms) WHERE {

  ?film dbo:starring dbr:Jennifer_Aniston .

}
```

Answer : 46

3. What is the last work she performed and when?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?film ?date WHERE {
  ?film dbo:starring dbr:Jennifer_Aniston ;
        dbo:releaseDate ?date .
}
ORDER BY DESC(?date)
LIMIT 1
```

Answer: The Morning Show

4. How old is her partner?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?birthDate WHERE {
  dbr:Brad_Pitt dbo:birthDate ?birthDate .
}
```

Answer : 1963-12-18

5. Give me the total number of actors that performed in the latest film of your favorite actor.

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT (COUNT(?actor) AS ?numActors) WHERE {
  ?film dbo:starring dbr:Jennifer_Aniston ;
        dbo:releaseDate ?date ;
        dbo:starring ?actor .
}
ORDER BY DESC(?date)
LIMIT 1
```

Answer : 16

6. Who directed the latest film?

```
SELECT ?director WHERE {
  ?film dbo:starring dbr:Jennifer_Aniston ;
        dbo:releaseDate ?date ;
        dbo:director ?director .
}
ORDER BY DESC(?date)
LIMIT 1
```

Answer : Rob_Schiller

7. What is the genre of her latest film?

```
SELECT ?genre WHERE {
  ?film dbo:starring dbr:Jennifer_Aniston ;
        dbo:releaseDate ?date ;
        dbo:genre ?genre .
}
ORDER BY DESC(?date)
LIMIT 1
```

Answer : Drama_(film_and_television)

8. In which country was the latest film produced?

```
SELECT ?country WHERE {
  ?film dbo:starring dbr:Jennifer_Aniston ;
        dbo:releaseDate ?date ;
        dbo:country ?country .
}
ORDER BY DESC(?date)
LIMIT 1
```

Answer: United_States

9. When is her birthday?

```
SELECT ?birthDate WHERE {
```

```
  dbr:Jennifer_Aniston dbo:birthDate ?birthDate .
}
```

Answer : 1969-02-11

10. Where was she born ?

```
SELECT ?birthPlace WHERE {
  dbr: Jennifer_Aniston dbo: birthPlace ?birthPlace.
}
```

Answer : Los_Angeles

11. What is the release date of The Morning Show?

```
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?date WHERE {
  <http://dbpedia.org/resource/The_Morning_Show_(American_TV_series)> dbo:releaseDate ?date .
}
```

Answer : 2019-11-01

12. Which actors co-starred with Jennifer Aniston in The Morning Show?

```
SELECT ?property ?actor WHERE {
  <http://dbpedia.org/resource/The_Morning_Show_(American_TV_series)> ?property ?actor .

  FILTER (
    regex(str(?property), "starring", "i")
```

```
  )
}
```

Answer :

Steve Carell

Mark Duplass

Reese Witherspoon

Julianna Margulies

13. List the television series in which Jennifer Aniston appeared.

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT DISTINCT ?series WHERE {
  ?series dbo:starring dbr:Jennifer_Aniston .
}
```

Answer :

Friends

The Morning show

Ferris_Bueller

Molloy

14. Who are the main actors in friends series?

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
PREFIX dbo: <http://dbpedia.org/ontology/>


SELECT ?actor WHERE {
  dbr:Friends dbo:starring ?actor .
}
```

Answer :

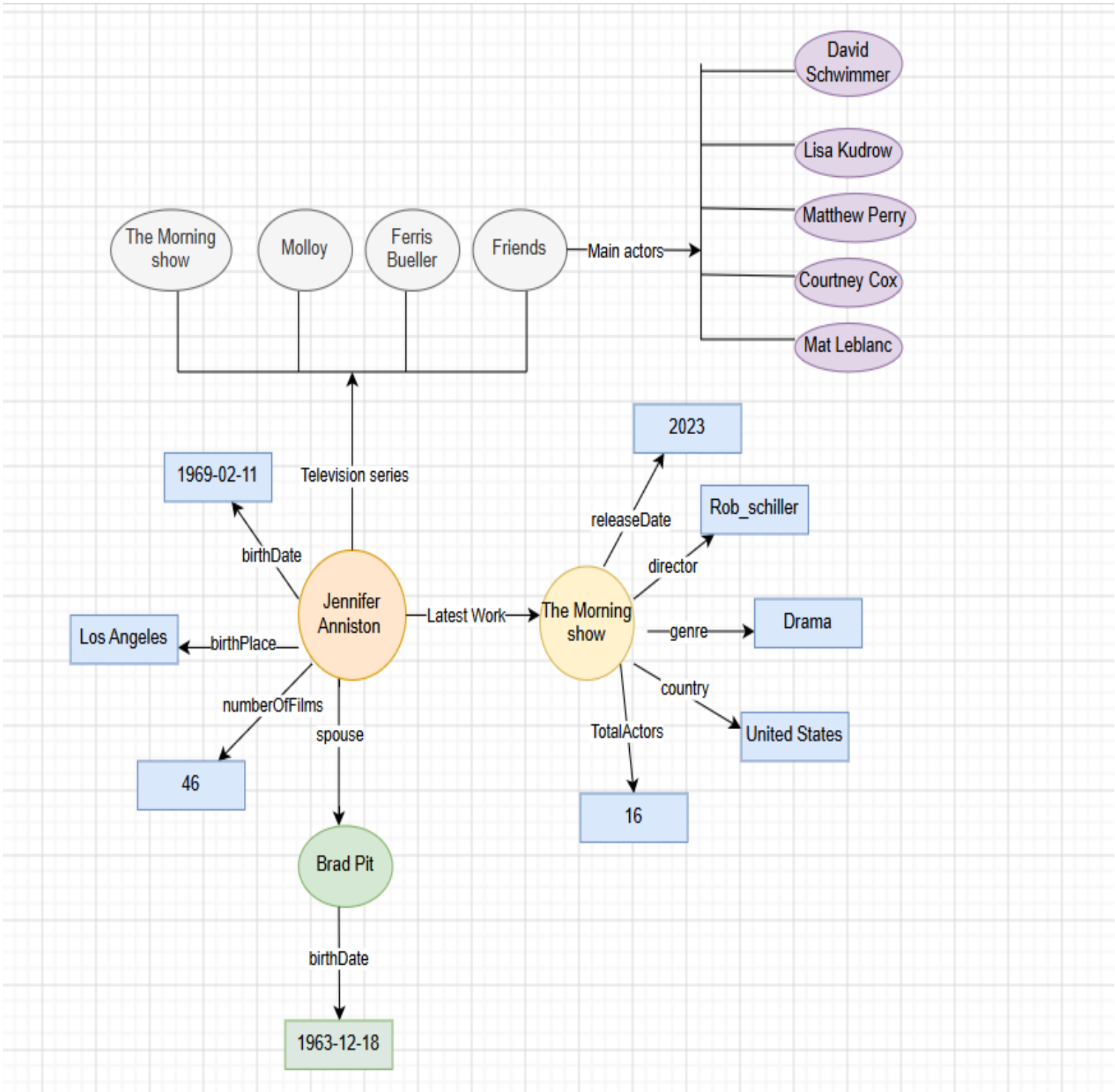
Lisa_Kudrow

Matthew_Perry

Courteney_Cox

Matt_LeBlanc

David_Schwimmer


David Schwimmer
Lisa Kudrow
Matthew Perry
Courtney Cox
Mat Leblanc
The Morning show
Molloy
Ferris Bueller
Friends
Main actors
2023
1969-02-11
Television series
releaseDate
Rob_schiller
birthDate
director
Jennifer Anniston
Latest Work
The Morning show
Los Angeles
birthPlace
genre
Drama
country
numberOfFilms
spouse
TotalActors
United States
46
16
Brad Pit
birthDate
1963-12-18